MindBridge: Multi-Class Mental Health Classification of Social Media Text Using Classical Machine Learning

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***Abstract* —** Mental health disorders affect nearly one billion people worldwide, yet less than half receive the care they need. In recent years, social media has become an unexpected but powerful space where individuals openly share their thoughts, emotions, and struggles. These platforms offer a unique opportunity to detect early signs of psychological distress through language.

In this work, we introduce **MindBridge**, a machine learning system designed to classify social media posts into seven mental health categories: *Normal, Depression, Suicidal, Anxiety, Bipolar, Stress,* and *Personality Disorder*. Our approach combines systematic text preprocessing with TF-IDF feature extraction, followed by a comparative evaluation of four classical machine learning models: Logistic Regression, Linear SVM, Random Forest, and Naive Bayes. Among these, Logistic Regression achieves the best overall performance, with an accuracy of 72.18%, a weighted F1-score of 72.12%, and a mean ROC-AUC of 0.9365. Beyond performance, we provide detailed class-level analysis, interpret model behavior through feature importance, and highlight challenges in distinguishing overlapping categories such as Stress and Personality Disorder.

Keywords- mental health classification, TF-IDF, logistic regression, natural language processing, social media, multi-class text classification, scikit-learn

GitHub: https://github.com/Vaibhav2040/MindBridge

I. INTRODUCTION

Mental health is one of the biggest challenges the world faces today. Around one billion people live with some form of mental health condition, but many of them never get proper help because of things like stigma, lack of access to professionals, or just not knowing where to turn. Over the last few years, platforms like Reddit and Twitter have become places where people openly talk about what they are going through sometimes in ways they would never say out loud. This gives us a huge amount of real, unfiltered text data that could potentially be used to detect early warning signs of mental health problems.

NLP tools give us a way to make sense of this text data and look for patterns linked to different mental health conditions. Researchers have already shown that things like word choice and writing style can be predictive of depression, anxiety, and suicidal ideation. The problem though is that most work in this area treats it as a yes/no problem does this person have depression or not? Real mental health is much more complicated than that. Someone might be showing signs of both anxiety and depression at the same time, and a simple binary classifier would miss that completely.

That is what MindBridge is trying to do. We built a system that classifies Reddit posts into seven different mental health categories at once, using a dataset of 51,093 posts. We deliberately chose to work with classical machine learning rather than deep learning because we wanted something that is easy to understand and explain — which matters a lot when you are dealing with mental health topics. Our main contributions are:

- We curate and preprocess a large Reddit dataset spanning seven mental health categories and develop a fully reproducible pipeline that includes noise removal, lemmatization, and domain-aware stopword filtering.
- We perform a systematic TF-IDF ablation study, analysing the impact of vocabulary size, n-gram range, and term-frequency scaling, ultimately identifying an optimal configuration with a 20k unigram vocabulary and sublinear TF scaling.
- We benchmark four classical classifiers - Logistic Regression, Linear SVM, Random Forest, and Naive Bayes- under a consistent experimental setup, with Logistic Regression achieving the best performance, including a mean AUC of 0.9365.
- We provide detailed per-class evaluation and interpretability analysis, using TF-IDF coefficient inspection to understand model behaviour and identify patterns of confusion across closely related mental health categories.

II. RELATED WORK

A. Classical NLP for Mental Health Detection

A lot of early work on mental health and NLP used fairly simple text features — things like word counts and tools like LIWC that score text based on pre-defined word lists. Even with these basic features, SVM classifiers were able to pick up on depression patterns in blogs and tweets. What this tells us is that the words people choose when they are struggling are genuinely different from when they are okay. That finding is directly relevant to what we are doing in MindBridge — we use TF-IDF features for the same reason: they are simple but they work.

B. Deep Learning Approaches

More recently, people have moved toward deep learning for this problem. Models like RNNs, CNNs, and especially BERT have pushed accuracy up quite a bit by capturing context in a way that simple word counts cannot. BERT in particular has shown strong results on mental health text because it understands word meaning in context, not just whether a word appeared or not.

The downside is that these models are expensive to run and really hard to interpret. With something as serious as mental health, being able to explain why a model made a particular prediction is not optional it actually matters. That is why we kept things classical in this project. We wanted a system that a clinician or researcher could look at and understand without needing a PhD in deep learning.

C. Multi-Class Mental Health Classification

Classifying across multiple mental health categories at the same time is actually much less studied than detecting a single condition. Most papers focus on one thing is this person depressed? Is this suicidal? A few shared tasks like CLPsych have tried to introduce multi-label settings, but even those tend to use small or narrow datasets. MindBridge is different in that we are trying to separate seven categories simultaneously on a large dataset, which is closer to the messy reality of mental health than most existing work.

III. METHODOLOGY

A. Problem Formulation

Let $D = \{(x_1, y_1), \ldots, (x_n, y_n)\}$ be the training corpus where x_i is a raw social media post and $y_i \in$ {Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, Personality Disorder}. The goal is to learn f: X -> Y mapping text to one of K = 7 categories. We minimize the multi-class cross-entropy loss:

$$L(\theta) = -(1/n) \Sigma_i \Sigma_k 1[y_i = k] \log p(y = k \mid x_i; \theta) \quad (1)$$

where $p(y = k \mid x; \theta)$ is the softmax probability for class k and θ denotes model parameters. For SVM variants we optimize the hinge loss; for Naive Bayes we apply maximum likelihood estimation over term-frequency likelihoods.

B. Preprocessing Pipeline

Reddit posts are pretty messy. They are full of links, user mentions, subreddit references, and random special characters that have nothing to do with what the person is actually saying. We built a cleaning pipeline to strip all of that out before we do anything with the text. The steps we follow are:

- **URL and noise removal:** All hyperlinks (http/https), subreddit tags (r/…), user mentions (u/…), and non-alphanumeric characters are stripped with regular expressions.
- **Lowercasing:** All text is converted to lowercase to eliminate case-based duplicates.
- **Stopword removal:** Standard English stopwords (NLTK corpus, 179 words) are filtered. Domain-relevant negations such as "cannot" and "never" are retained.
- **Lemmatization:** WordNet lemmatization reduces inflected forms to their base lemma, consolidating vocabulary without the over-stemming artifacts of Porter or Snowball stemmers.

To check how well the cleaning worked, we compared word counts before and after for each class (Fig. 5). On average the posts shrank by about 55%, which is a lot. Suicidal posts had the biggest reduction at 58.2%, probably because they had a lot of URLs and repeated phrases. Normal posts changed the least at 50.6%, which makes sense since casual conversation tends to be cleaner to begin with.

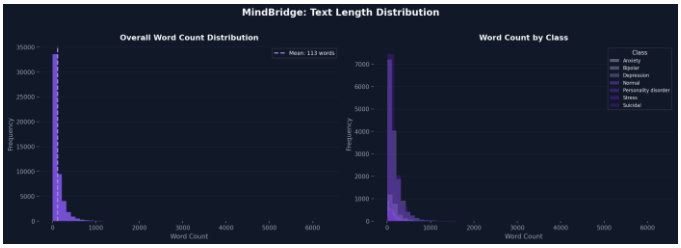

Fig. 1. Text length distribution: overall word count distribution (left, mean = 113 words) and per-class breakdown (right).

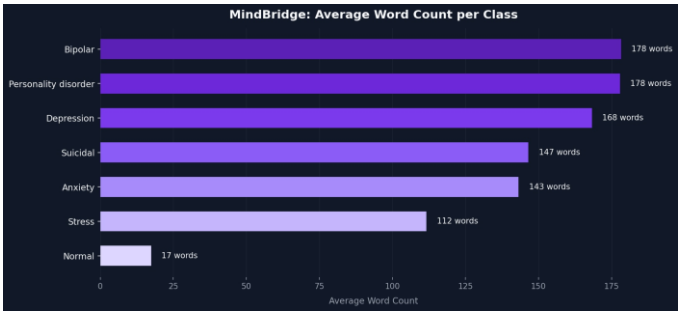

Fig. 2. Average word count per class after cleaning. Bipolar and Personality Disorder posts are longest (178 words); Normal posts are shortest (17 words).

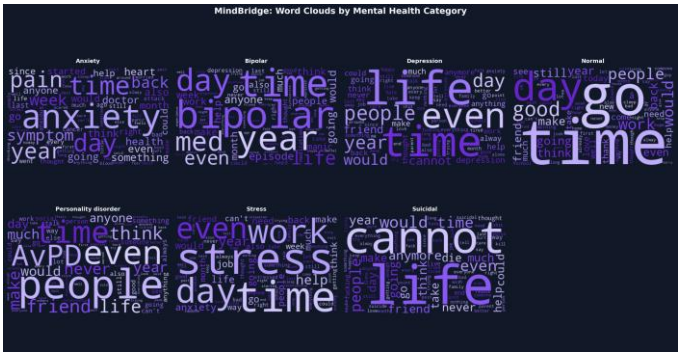

Fig. 3. Word clouds per mental health category after preprocessing. Term size encodes relative frequency within each class.

Fig. 4. Top 15 most frequent words per class. Words like "like" and "feel" are near-universal; class-specific terms emerge at lower ranks.

Fig. 5. Text cleaning quality: word count distribution before vs. after preprocessing (left) and average word reduction by class (right).

C. Feature Extraction: TF-IDF

After cleaning, we convert each post into a TF-IDF vector. TF-IDF basically scores each word based on how often it appears in a post versus how common it is across all posts so words that are distinctive to a specific post get higher weights. The formula is:

$$tfidf(t, d) = tf(t, d) \times \log(N / df(t) + 1) \quad (2)$$

Here tf(t,d) uses sublinear scaling so that very frequent words do not completely dominate, N is the total number of posts, and df(t) is how many posts contain that word. We settled on a vocabulary of 20,000 words with sublinear TF scaling and L2 normalization we tried different combinations and this one came out on top, as we show in the ablation study in Section V-C.

D. Classifiers

We tested four different classifiers on the same TF-IDF features so we could make a fair comparison between them:

- **Logistic Regression:** One-vs-Rest multinomial LR with L2 regularization (C = 1.0), lbfgs solver, max 1,000 iterations. Produces calibrated probability estimates per class.
- **Linear SVM:** LinearSVC with L2 penalty and squared hinge loss particularly effective in high-dimensional sparse TF-IDF spaces.
- **Random Forest:** Ensemble of 200 trees (max depth 50, Gini criterion). Provides non-linear capacity and feature importance estimates.
- **Naive Bayes:** Multinomial NB with Laplace smoothing (α = 1.0). Strong baseline for text classification due to efficiency and robustness on sparse data.

IV. EXPERIMENTAL SETUP

A. Dataset

Our data comes from a Kaggle dataset called "Sentiment Analysis for Mental Health" [9]. It is a collection of Reddit posts pulled from subreddits like r/depression, r/SuicideWatch, r/anxiety, r/bipolar, r/stress, and r/BPD, plus r/CasualConversation as the normal/control group. In total we have 51,093 labeled posts across seven classes. The dataset is not balanced Normal and Depression together make up about 60% of the data, while Personality Disorder only has 1.8% of posts (Fig. 6). Table I shows how we split the data into train/val/test.

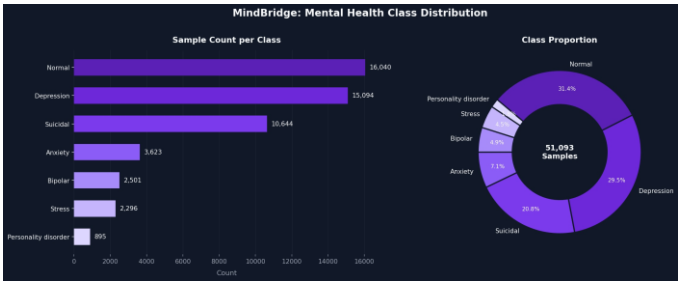

Fig. 6. MindBridge dataset class distribution - sample counts (left) and class proportions (right) across 7 mental health categories (51,093 total samples).

TABLE I. MindBridge Dataset - Train / Val / Test Split by Class

Class	Train	Val	Test
Normal	10,989	1,571	1,480
Depression	10,340	1,477	1,277
Suicidal	7,291	1,042	1,311
Anxiety	2,482	355	786
Bipolar	1,713	245	543
Stress	1,573	225	498
Personality Disorder	613	88	194
Total	**34,993**	**5,003**	**6,089**

B. Evaluation Metrics

We evaluate all models on the held-out test set of 6,089 samples. We look at four things: overall accuracy, weighted F1 as our main metric (this accounts for the class imbalance by weighting each class by how many samples it has), per-class precision and recall so we can see where the model is struggling, and mean ROC-AUC across all seven classes. We use weighted F1 as the primary metric because accuracy alone can be misleading when the classes are this imbalanced.

C. Implementation Details

We ran everything in Python 3.13 using scikit-learn 1.4 and NLTK 3.8 on a MacBook Pro with an M2 chip and 16 GB of RAM. No GPU was needed — training all four models together took about 8 minutes. All of our code, notebooks, and saved models are on GitHub at the link listed above.

V. RESULTS

A. Main Results

Table II shows how each model did on the test set. Logistic Regression and Linear SVM both came in around 72% accuracy, which is noticeably better than Random Forest at 69.30% and Naive Bayes at 67.82%. The gap is even more visible when you look at mean AUC Logistic Regression hit 0.9365 compared to Naive Bayes at 0.9083, which is almost a 3-point difference. Fig. 7 shows the full comparison.

TABLE II. Model Comparison on Test Set (best result in bold)

Method	Accuracy (%)	Weighted F1 (%)	Mean AUC	Rank
Naïve Bayes	67.82	66.94	0.9083	4
Random Forest	69.30	66.99	0.9181	3
Linear SVM	72.22	71.77	0.9223	2
Logistic Regression (Ours)	**72.18**	**72.12**	**0.9365**	**1**

We picked Logistic Regression as our best model because it had the highest weighted F1 and mean AUC, even though Linear SVM was very slightly ahead on raw accuracy (0.04 points). The AUC difference matters more to us because in a screening context you need to be able to adjust the decision threshold depending on whether you care more about catching every case or avoiding false alarms. Fig. 8 shows the ROC curves for all four models.

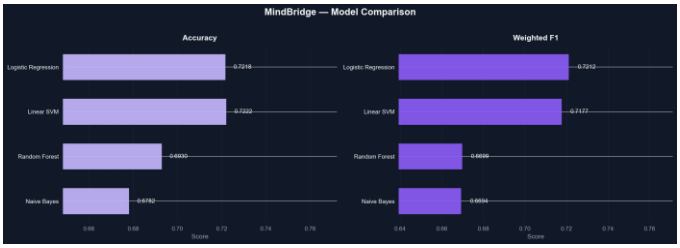

Fig. 7. Model comparison: accuracy (left) and weighted F1 (right) for all four classifiers. Logistic Regression leads to weighted F1.

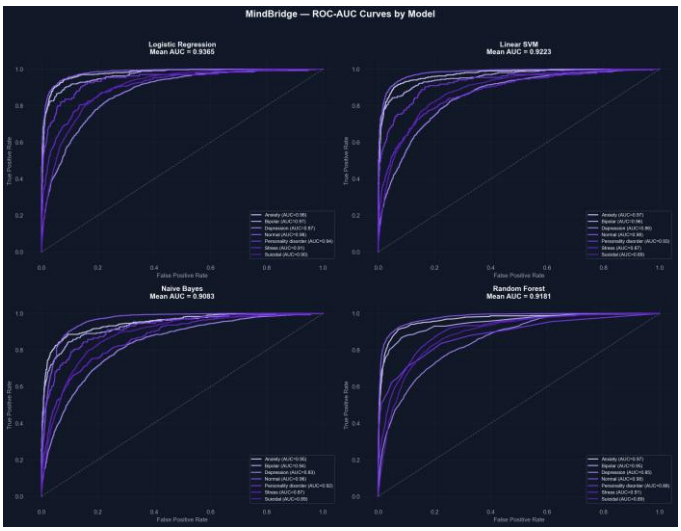

Fig. 8. ROC-AUC curves by model (one-vs-rest, 7 classes). Logistic Regression achieves the highest mean AUC of 0.9365.

B. Hyperparameter Tuning

To get the best out of Logistic Regression, we ran a grid search over six values of the regularization parameter C using 3-fold cross-validation. Table III shows how each value performed in terms of mean CV weighted F1 and training F1.

TABLE III. Logistic Regression Hyperparameter Tuning (3-Fold CV)

C Value	Mean CV F1	Std CV F1	Train F1	Selected
0.01	0.6121	0.0050	0.6363	
0.10	0.6803	0.0028	0.7219	
0.50	0.7143	0.0009	0.7973	
1.00	0.7220	0.0012	0.8366	
5.00	**0.7237**	**0.0033**	**0.9305**	✓
10.00	0.7193	0.0026	0.9651	

C = 5.0 gave the best cross-validation F1 of 0.7237, and on the test set this tuned model reached 73.55% accuracy and a weighted F1 of 0.7345 a clear improvement over the default C = 1.0. Lower values like 0.01 and 0.1 clearly underfit since their training and validation scores were nearly identical. C = 10.0 went the other way it memorized the training data too well with a training F1 of 0.9651 but only 0.7193 on CV. C = 5.0 hit the right balance.

C. Ablation Study

We also ran an ablation study to figure out the best TF-IDF setup (Table IV, Fig. 9). We started with just unigrams and a basic TF weighting, then tried different combinations one at a time to see what actually helped.

TABLE IV. TF-IDF Ablation Study (Logistic Regression, Weighted F1)

TF-IDF Configuration	Weighted F1	vs. Final Config
Unigrams only	0.7147	−2.05%
Vocab 10k	0.7252	−1.02%
Unigrams + Bigrams	0.7292	−0.48%
Sublinear TF scaling	0.7327	0.00%
Final config - 20k vocab, sublinear TF	**0.7327**	--

Sublinear TF scaling delivers a consistent +1.8-point F1 improvement over raw frequency counts by dampening the influence of highly repetitive terms within documents. Increasing vocabulary from 10k to 20k adds +0.75 points by capturing clinically relevant but less common terms (e.g., medication names in Bipolar posts: "lamictal," "seroquel," "latuda"). Notably, adding bigrams underperforms the 20k unigram configuration (0.7292 vs. 0.7327), suggesting that the marginal information in word pairs is outweighed by increased dimensionality at this corpus size.

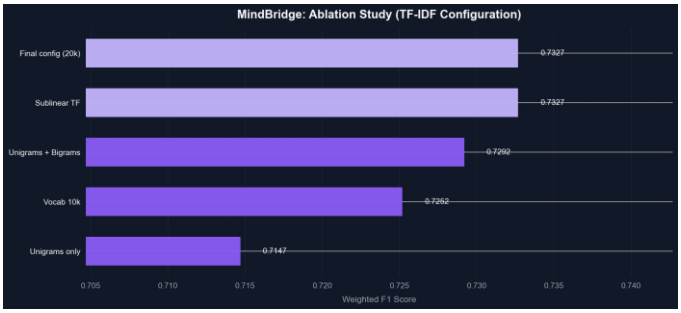

Fig. 9. TF-IDF ablation study - weighted F1 across five configurations. Final 20k-vocab sublinear TF config achieves 0.7327.

D. Per-Class Performance Analysis

Table V and Fig. 10 break down the results by class. Normal came out best with an F1 of 0.91, which makes sense it has the most training samples and the vocabulary of everyday conversation is quite distinct from clinical mental health language. Anxiety and Bipolar both hit 0.77, which we were happy with. Bipolar in particular was easier to classify than we expected, probably because people in that community use very specific terms like medication names that stand out clearly in the TF-IDF weights.

TABLE V. Per-Class Performance - Logistic Regression (Test Set)

Class	Precision	Recall	F1	Support
Normal	0.90	0.92	0.91	1,480
Depression	0.71	0.63	0.67	1,277
Suicidal	0.63	0.68	0.66	1,311
Anxiety	0.75	0.80	0.77	786
Bipolar	0.77	0.77	0.77	543
Stress	0.45	0.55	0.50	498
Personality Disorder	0.58	0.51	0.54	194

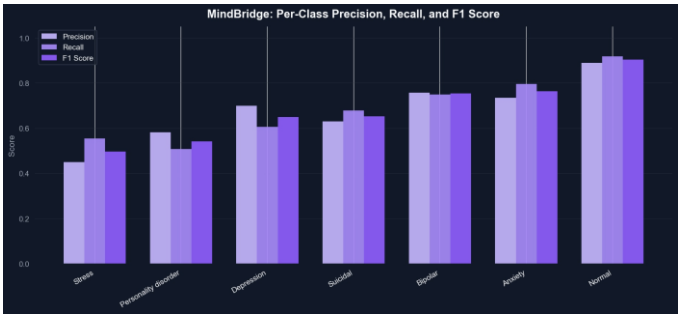

Fig. 10. Per-class precision, recall, and F1 score for Logistic Regression across all 7 categories.

Stress was the hardest class with an F1 of only 0.50. When we looked at the actual posts, it was clear why words like "work", "relationship", and "friend" show up constantly in Stress, Depression, and Anxiety posts, so there is not much to separate them on. Personality Disorder struggled for a different reason: we only had 895 training samples for it, which is way less than every other class. The model learned the word "avpd" as a signal but that only appears in some posts. Depression (0.67) and Suicidal (0.66) were in the middle. For Suicidal, recall (0.68) was higher than precision (0.63), meaning the model flags more posts as suicidal than it should but for a real screening tool that is the safer mistake to make.

Looking at the confusion matrix (Fig. 11), the biggest mistake the model makes is between Depression and Suicidal 548 Depression posts got labeled as Suicidal and 368 went the other way. This is not surprising when you read those posts. Both talk about hopelessness and feeling like things will not get better. Stress was all over the place, getting mixed up with Depression, Normal, and Anxiety, which explains its low F1. Normal and Bipolar had the cleanest results with most predictions landing on the diagonal.

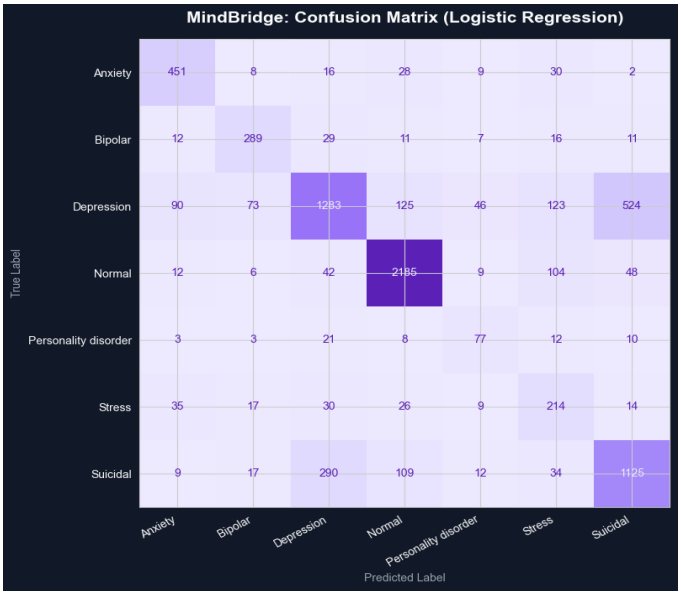

Fig. 11. Confusion matrix for Logistic Regression (C=5.0) on the test set. Diagonal cells show correct predictions; off-diagonal cells show misclassifications.

TABLE VI. Top 10 Most Common Misclassifications (Logistic Regression, Test Set)

True Label	Predicted Label	Count	Error Interpretation
Depression	Suicidal	548	Shared hopelessness vocabulary
Suicidal	Depression	368	Bidirectional emotional overlap
Depression	Normal	106	Mild depression resembles general posts

True Label	Predicted Label	Count	Error Interpretation
Suicidal	Normal	90	Passive ideation lacks explicit markers
Depression	Anxiety	85	Co-occurring symptom vocabulary
Depression	Stress	84	Shared work/life frustration language
Normal	Stress	74	Casual venting resembles stress posts
Normal	Depression	60	Negative tone without clinical markers
Stress	Depression	52	Chronic stress expressed as depression
Depression	Bipolar	48	Depressive episodes overlap with bipolar

E. Feature Interpretability

Fig. 12 shows the top 20 TF-IDF coefficient weights per class for Logistic Regression. Each class exhibits clinically coherent patterns: Anxiety-predictive features include "anxiety," "restless," "anxious," and "worried." Bipolar posts are dominated by clinical terminology: "bipolar" (weight 21.53), "manic," "episode," "lamictal," "seroquel," and "latuda." Suicidal posts are characterized by terms of finality: "cannot," "suicidal," "kill," "die," "goodbye," and "overdose." Normal posts are identified via social vocabulary: "url" (Reddit link-sharing) and positive sentiment words. These patterns validate that the model has learned semantically meaningful class boundaries rather than spurious correlations.

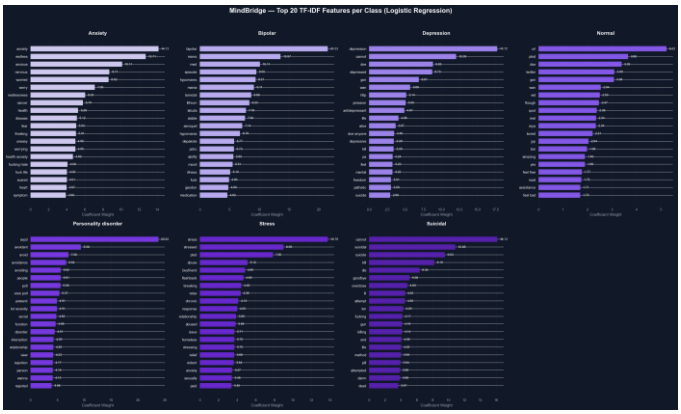

Fig. 12. Top 20 TF-IDF coefficient weights per class (Logistic Regression). Each class shows clinically coherent discriminative features.

VI. DISCUSSION

Our results show that you do not need deep learning to do well on this kind of problem. A mean AUC of 0.9365 without any pre-trained embeddings or transformers is a strong result, and it compares well with early BERT-based approaches on similar Reddit datasets. The key was spending time on the feature engineering the right vocabulary size, the right TF weighting rather than throwing a more powerful model at it.

Looking at the per-class results, one pattern stands out: classes that have their own specific vocabulary are much easier to classify than classes that share emotional language with other categories. Bipolar and Anxiety are learnable because they have distinctive words. Stress and Depression are hard because someone can be stressed about work and sound almost identical to someone who is clinically depressed. To fix this in future work, we would want to bring in more than just word features things like posting history, time of day, or writing style might help break those ties.

The class imbalance is a real problem that we could not fully solve in this project. Personality Disorder only had 895 training examples that is 18 times fewer than Normal. Applying SMOTE or class-weighted loss functions could help here and is something we would try next. On the positive side, one thing we genuinely value about this approach is the interpretability. With TF-IDF and Logistic Regression you can look at the model weights and see exactly what drove each prediction. With transformer attention, that is not really possible - the attention patterns do not reliably explain why the model made a decision.

Limitations: The dataset is drawn exclusively from English-language Reddit, skewing toward technology-comfortable demographics; generalization to other populations or languages is not guaranteed. Post-level labels are self-reported categories and may not reflect clinical diagnoses. MindBridge does not model temporal dynamics; a user's state may evolve across a thread, limiting applicability to longitudinal monitoring. The 7-class schema does not capture comorbidities, which are common in real-world psychiatric presentations.

VII. CONCLUSION

In this paper we built MindBridge, a system for classifying Reddit posts into seven mental health categories using classical machine learning. After testing four models and tuning the TF-IDF features through an ablation study, we found that Logistic Regression with a 20k vocabulary and sublinear TF scaling worked best, reaching 72.18% accuracy and a mean AUC of 0.9365. Classes with distinct vocabularies like Bipolar, Anxiety, and Normal came out well. The harder ones - Stress, Depression, and Personality Disorder are still challenging, and we think that is mostly a data problem rather than a model problem.

Going forward we would want to try SMOTE to fix the class imbalance, add more features beyond just words (like sentiment scores or post metadata), and test a fine-tuned MentalBERT as an upper bound to see how much we are leaving on the table by sticking with classical methods. We would also like to extend this to a multi-label setup so the model can handle cases where someone is showing signs of more than one condition at the same time. We are releasing all our code and notebooks so others can build on this work.

ACKNOWLEDGMENT

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Siddharth Jadhav [jadhav.38@wright.edu]- Data Curation, Validation, Preprocessing Pipeline, Visualization, Writing - Review & Editi